

Polynomial compression for fixed-fold sumset sizes

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Abstract

For fixed h , let $N(h, k)$ be the least N such that every cardinality $|hA|$ attained by a k -element set of integers is also attained by a k -element subset of $\{1, \dots, N\}$. We prove, using Rajagopal's fixed- h theorem on the range of possible values of $|hA|$, that

$$N(h, k) \leq k^{C_h}$$

for every fixed h . The new ingredient is a polynomial-diameter replacement for the sparse geometric blocks in Rajagopal's large-value construction. The replacement is carried out at the level of the low-degree Hilbert functions, which are the only data relevant to h -fold sumsets when h is fixed.

1 Introduction

For a finite set $A \subset \mathbb{Z}$ and a positive integer h , write

$$hA = \{a_1 + \dots + a_h : a_i \in A\}$$

with repetitions allowed. Define

$$\mathcal{R}(h, k) = \{|hA| : A \subset \mathbb{Z}, |A| = k\}.$$

Let $N(h, k)$ be the least positive integer N such that

$$\mathcal{R}(h, k) = \{|hA| : A \subseteq \{1, \dots, N\}, |A| = k\}.$$

Equivalently, one may work in intervals $[0, N - 1]$ and translate at the end.

The purpose of this note is to prove the following polynomial compression bound.

Theorem 1.1. *For every fixed $h \geq 1$ there is a constant C_h such that, for all $k \geq 2$,*

$$N(h, k) \leq k^{C_h}.$$

The proof uses Rajagopal's theorem on the range of possible fixed-fold sumset sizes. Put

$$M_{h,k} = hk - h + 1, \quad K_{h,k} = \binom{h+k-1}{h},$$

and define

$$\Delta_{h,k} = \bigcup_{\ell=0}^{\min(h,k)-3} [M_{h,k} + \ell h + 1, M_{h,k} + \ell h + (h - 2 - \ell)].$$

Here and below, $[u, v]$ denotes the set of integers n with $u \leq n \leq v$; if $u > v$ the interval is empty.